

● HARD

● GEOMETRY AND TRIGONOMETRY

Area and volume

02

10 questions · ~15 min

Q1

GRID-IN

GEOMETRY AND TRIGONOMETRY

A right circular cone has a height of 22 centimeters cm and a base with a diameter of 6 cm. The volume of this cone is $n\pi$ cm³. What is the value of n ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Rectangles ABCD and EFGH are similar. The length of each side of EFGH is 6 times the length of the corresponding side of ABCD. The area of ABCD is 54 square units. What is the area, in square units, of EFGH?

- A. 9
- B. 36
- C. 324
- D. 1,944

The volume of right circular cylinder A is 22 cubic centimeters. What is the volume, in cubic centimeters, of a right circular cylinder with twice the radius and half the height of cylinder A?

- A. 11
- B. 22
- C. 44
- D. 66

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY

A right square prism has a height of 14 units. The volume of the prism is 2,016 cubic units. What is the length, in units, of an edge of the base?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A cube has an edge length of 68 inches. A solid sphere with a radius of 34 inches is inside the cube, such that the sphere touches the center of each face of the cube. To the nearest cubic inch, what is the volume of the space in the cube not taken up by the sphere?

- A. 149,796
- B. 164,500
- C. 190,955
- D. 310,800

Q6

GRID-IN

GEOMETRY AND TRIGONOMETRY

A right rectangular prism has a length of 28 centimeters cm, a width of 15 cm, and a height of 16 cm. What is the surface area, in cm^2 , of the right rectangular prism?

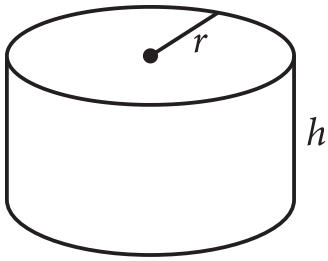
STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The floor of a ballroom has an area of 600 square meters. An architect creates a scale model of the floor of the ballroom, where the length of each side of the model is $\frac{1}{10}$ times the length of the corresponding side of the actual floor of the ballroom. What is the area, in square meters, of the scale model?

- A. 6
- B. 10
- C. 60
- D. 150

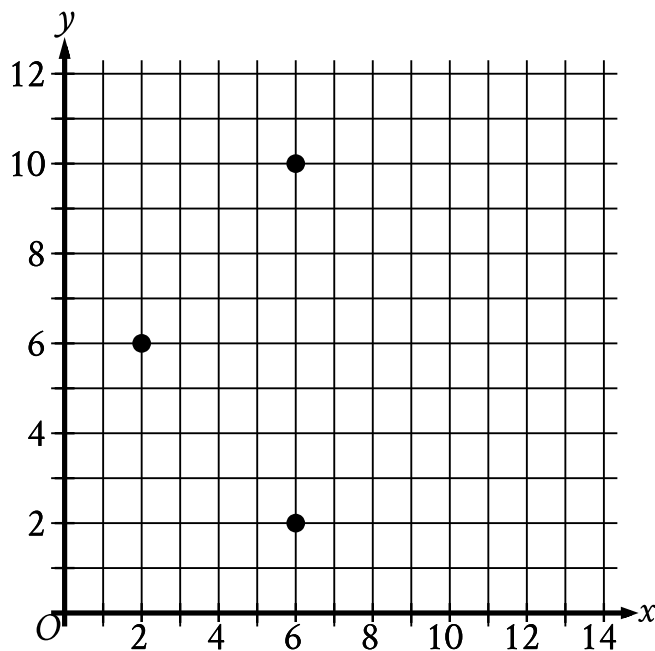


The figure shown is a right circular cylinder with a radius of r and height of h . A second right circular cylinder (not shown) has a volume that is 392 times as large as the volume of the cylinder shown. Which of the following could represent the radius R , in terms of r , and the height H , in terms of h , of the second cylinder?

- A. $R = 8r$ and $H = 7h$
- B. $R = 8r$ and $H = 49h$
- C. $R = 7r$ and $H = 8h$
- D. $R = 49r$ and $H = 8h$

A manufacturer determined that right cylindrical containers with a height that is 4 inches longer than the radius offer the optimal number of containers to be displayed on a shelf. Which of the following expresses the volume, V , in cubic inches, of such containers, where r is the radius, in inches?

- A. $V = 4\pi^3$
- B. $V = \pi(2r)^3$
- C. $V = \pi r^2 + 4\pi r$
- D. $V = \pi r^3 + 4\pi r^2$



- From left to right, the 3 points have the following coordinates:
 - (2 comma 6)
 - (6 comma 10)
 - (6 comma 2)

The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
(0)	(0)	(0)	(0)
(1)	(1)	(1)	(1)
(2)	(2)	(2)	(2)
(3)	(3)	(3)	(3)
(4)	(4)	(4)	(4)
(5)	(5)	(5)	(5)

6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.